Review of some real analysis

Spaces of continuous functions

Given $a, b \in \mathbb{R}$ w/ $a<b$, define

$$C^0([a,b],\mathbb{R}^n) = \{ f:[a,b]\to\mathbb{R}^n \mid f \text{ is continuous}\}.$$

More generally, could also consider $A \subset \mathbb{R}^m$,

$$C^0(A,\mathbb{R}^n) = \{ f: A \to \mathbb{R}^n \mid f \text{ is continuous}\}.$$

Fact $C^0([a,b],\mathbb{R}^n)$ is a real vector space w/ addition defined by $(f+g)(x) = f(x)+g(x)$, and scalar mult defined by $(cf)(x) = cf(x)$.

Def Let V be a real vector space. A **norm** on V is a map $\|\cdot\|: V \to [0,\infty)$ s.t.

(1) $\|x\| = 0 \iff x = 0$.

(2) $\forall c \in \mathbb{R},\ x \in V$: $\|cx\| = |c|\,\|x\|$

(3) $\forall x, y \in V$: $\|x+y\| \leq \|x\| + \|y\|$.

Def Fix $a, b \in \mathbb{R}$ w/ $a < b$, $n \in \mathbb{N}$. The **supremum (sup) norm** on $C^0([a,b], \mathbb{R}^n)$ is defined by

$$\|f\|_\infty = \sup_{t \in [a,b]} \|f(t)\|.$$

Prop The sup norm is a norm on $C^0([a,b],\mathbb{R}^n)$ for $a<b$, $n\in\mathbb{N}$.

pf $\|\cdot\|_\infty$ is clearly a map taking values in $[0,\infty)$. ✓

Next, WWTS that $\|f\|_\infty=0 \Leftrightarrow f=0$.

$\Leftarrow$) If $f=0$, then $\|f(t)\|=0$ $\forall t$, so $\|f\|_\infty=0$. ✓

$\Rightarrow$) Assume $\|f\|_\infty=0$. If $f\neq 0$, then $\exists\ t\in[a,b]$ s.t. $f(t)\neq 0$. Thus $\|f(t)\|>0$. Then $\|f\|_\infty \geq \|f(t)\|>0$, contradicting $\|f\|_\infty=0$. ✓

Next, WWTS $\|cf\|_\infty=|c|\,\|f\|_\infty$. Have

$$\|cf\|_\infty=\sup_{t\in[a,b]}\|cf(t)\|=\sup_{t\in[a,b]}|c|\,\|f(t)\|=|c|\sup_{t\in[a,b]}\|f(t)\|$$
$$=|c|\,\|f\|_\infty. ✓$$

Finally, WWTS that $\|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty$.

$$\|f+g\|_\infty = \sup_{t\in[a,b]} \|f(t)+g(t)\| \le \sup_{t\in[a,b]} \left(\|f(t)\|+\|g(t)\|\right)$$

$$\le \sup_{\substack{t\in[a,b]\\ s\in[a,b]}} \left(\|f(t)\|+\|g(s)\|\right)$$

$$= \sup_{t\in[a,b]} \|f(t)\| + \sup_{t\in[a,b]} \|g(t)\| = \|f\|_\infty + \|g\|_\infty. \quad \square$$

Def Given a set X, a **metric** on X is a function $d: X\times X \to [0,\infty)$ s.t.

(1) $d(x,y)=0 \Leftrightarrow x=y$;

(2) $d(x,y)=d(y,x)$ $\forall x,y\in X$;

(3) $d(x,y)\le d(x,z)+d(z,y)$ $\forall x,y,z\in X$.

Given a metric d on X, say (X,d) (or by abuse of terminology, X) is a **metric space**.

Every norm determines a metric

Let $(V, \|\cdot\|)$ be a normed vector space. Then (V, d) is a metric space w/ $d(x,y) := \|x-y\|$. (Exercise: check)

Convergence concepts for $C^0([a,b], \mathbb{R}^n)$

Def Let (f_k) $\left(= (f_k)_{k\in\mathbb{N}} = (f_k)_{k=1}^{\infty}\right)$ be a sequence of functions in $C^0([a,b], \mathbb{R}^n)$. Let $f: [a,b] \to \mathbb{R}^n$. We say that f_k converges to f $(f_k \to f)$

(1) pointwise if $\forall t \in [a,b]: f_k(t) \to f(t)$;

(2) uniformly if $\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall k \geq N : \forall t \in [a,b]:$
$$\|f_k(t) - f(t)\| < \varepsilon.$$

Note $f_k \to f$ ptwise if $\forall t \in [a,b] : \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall k \geq N :$
$$\|f_k(t) - f(t)\| < \varepsilon.$$

Fact $[f_k \to f \text{ uniformly}] \Leftrightarrow [\|f_k - f\|_\infty \to 0]$

Ex Consider the sequence (f_k) of functions $f_k : [0,\infty) \longrightarrow \mathbb{R}$ defined by

$$f_k(x) = \frac{\sin(kx)}{1+kx}$$

Q1 Does (f_k) converge ptwise?

Q2 Does (f_k) converge uniformly?

First note $f_k(0) = 0 \ \forall k$.

Next note that, $\forall x > 0$, $\lim\limits_{k \to 0} f_k(x) = 0$.

$\Rightarrow$ **A1** $f_k \to 0$ ptwise.

but **A2** (f_k) does NOT converge uniformly.

To see why, consider the sequence (x_k) in $[0,\infty)$ given by $x_k = 1/k$, and evaluate f_k on x_k. Then $\exists\, \varepsilon > 0$ s.t.

$$f_k(x_k) = \frac{\sin(k\frac{1}{k})}{1+k\frac{1}{k}} = \frac{\sin(1)}{2} > \varepsilon,$$

showing that $f_k \not\to 0$ uniformly.

<u>Def</u> A sequence (x_k) in a metric space (X,d) is **Cauchy** if $\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall k,l > N : d(x_k, x_l) < \varepsilon$.

<u>Def</u> A metric space (X,d) is **complete** if any Cauchy sequence in (X,d) converges (to some $x \in X$).

<u>Ex</u> $\mathbb{R}$ w/ standard metric $d(x,y) = |x-y|$ is complete.

Ex $\mathbb{R}\setminus\{0\}$ w/ standard metric is NOT complete;

e.g., $(x_k)_k$ def by $x_k = \frac{1}{k}$ is Cauchy,

but (x_k) does not converge in $\mathbb{R}\setminus\{0\}$.

Def Let $(V, \|\cdot\|)$ be a normed vector space. If the metric $d(x,y) = \|x-y\|$ on V is complete, then $(V, \|\cdot\|)$ is a Banach space.

Theorem Let $a, b \in \mathbb{R}$ w/ $a<b$, and $n \in \mathbb{N}$. The function space $C^0([a,b], \mathbb{R}^n)$ equipped w/ the sup metric $d(f,g) := \|f-g\|_\infty$ is a complete metric space, so $C^0([a,b], \mathbb{R}^n)$ w/ sup norm $\|\cdot\|_\infty$ is a Banach space.

Pf Let (f_n) be a Cauchy sequence in $C^0([a,b], \mathbb{R}^n)$.

Since $(f_n(t))$ is Cauchy $\forall t \in [a,b]$ and $\mathbb{R}^n$ is complete, $\exists f:[a,b] \to \mathbb{R}^n$ s.t. $f_n \to f$ ptwise. Fix any $\varepsilon > 0$. Since (f_n) is Cauchy, $\exists N > 0$ s.t. $\forall m,n > N$, $t \in [a,b]$, $|f_m(t) - f_n(t)| < \frac{\varepsilon}{2}$.

Hence for any $t \in [a,b]$ and $m,n > N$,

$$|f_n(t) - f(t)| < |f_n(t) - f_m(t)| + |f_m(t) - f(t)|$$
$$< \frac{\varepsilon}{2} + |f_m(t) - f(t)|.$$

Since $f_m(t) \to f(t)$, we may further choose m large enough (depending on t) s.t. $|f_m(t) - f(t)| < \frac{\varepsilon}{2}$ and hence $|f_n(t) - f(t)| < \varepsilon$ $\forall n > N$.

Since our choice of N did not depend on t, this shows $f_n \to f$ uniformly.

It remains only to show that f is continuous, so that $f \in C^0([a,b], \mathbb{R}^n)$. Fix $\varepsilon > 0$ and $s \in [a,b]$. Since $f_n \to f$ uniformly and (f_n) is Cauchy, $\exists N > 0$ s.t., $\forall m,n \geq N$ and $t \in [a,b]$, $|f_m(t) - f_n(t)| < \frac{\varepsilon}{3}$ and $|f_n(t) - f(t)| < \frac{\varepsilon}{3}$. Since f_N is cont at s, $\exists \delta > 0$ s.t. $|s-t| < \delta \Rightarrow |f_N(t) - f_N(s)| < \frac{\varepsilon}{3}$. Thus, $\forall t \in [a,b]$ s.t $|s-t| < \delta$, we have

$$|f(t) - f(s)| \leq |f(t) - f_N(t)| + |f_N(t) - f_N(s)| + |f_N(s) - f(s)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \quad \square$$

(Banach) Contraction mapping theorem

Let (X,d) be a complete metric space. Let $T: X \to X$ be a contraction map, i.e., $\exists\, c \in [0,1)$ s.t.

$$d(T(x), T(y)) \leq c\, d(x,y) \qquad \forall x,y \in X.$$

Then $\exists!\ x_* \in X$ s.t. $T(x_*) = x_*$. Moreover, $\forall x \in X$, (x_n) def by $x_n = T^n(x) = \underbrace{T \circ T \circ \cdots \circ T}_{n \text{ times}}(x)$

satisfies $x_n \to x_*$.